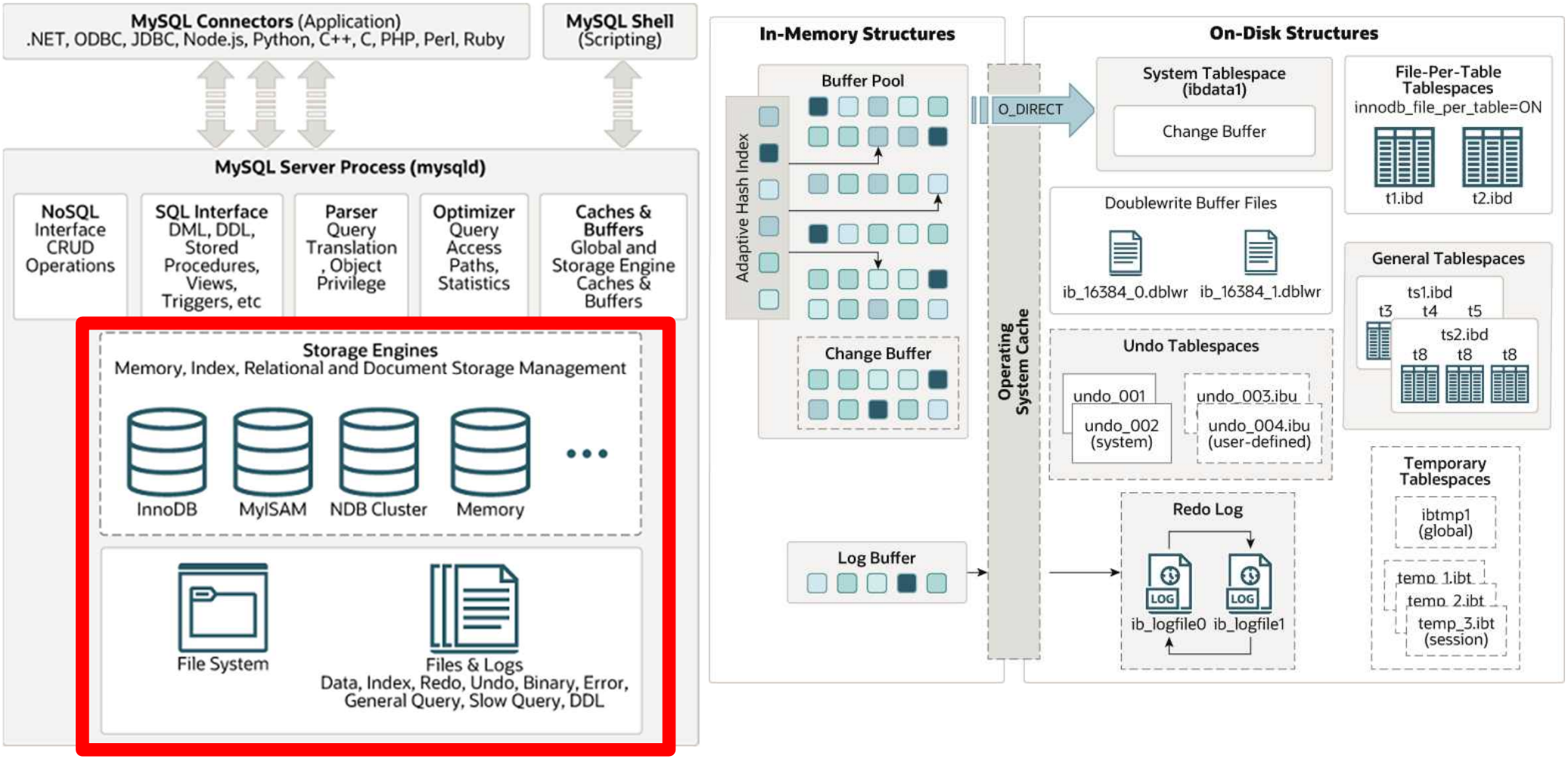
Physical Storage Systems

Seoul National University of Science and Technology

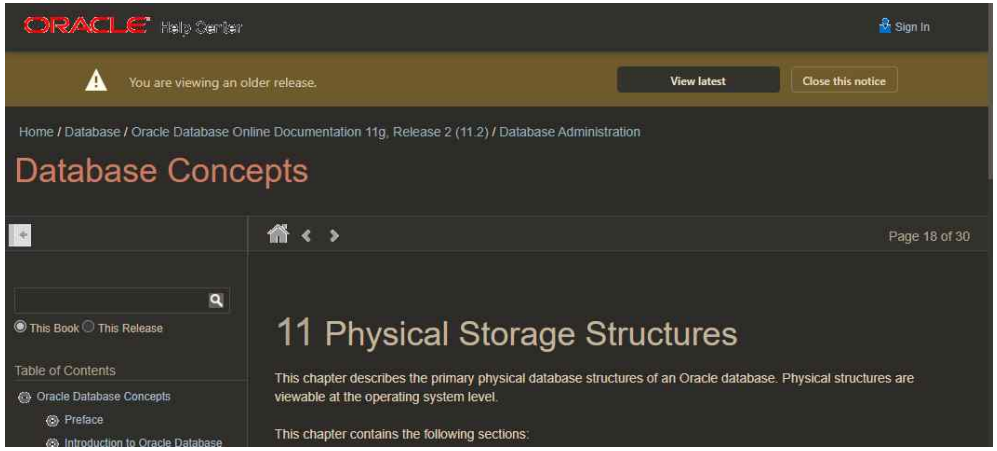
Database is essentially data storage system

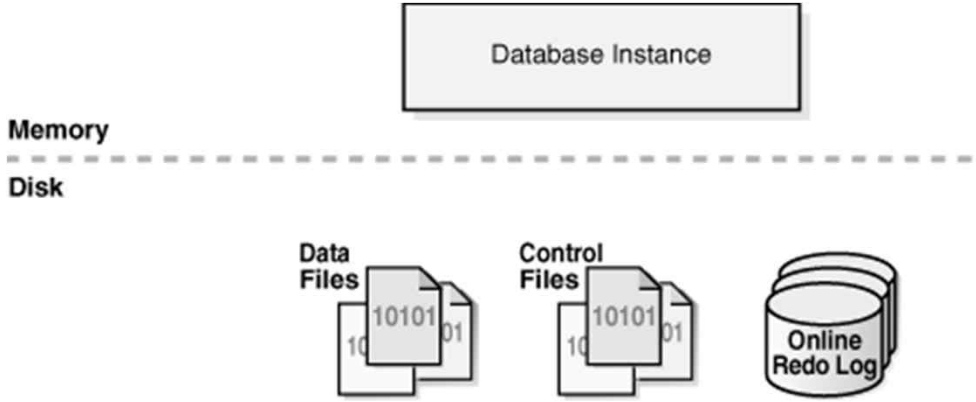
- MySQL and InnoDB storage engine

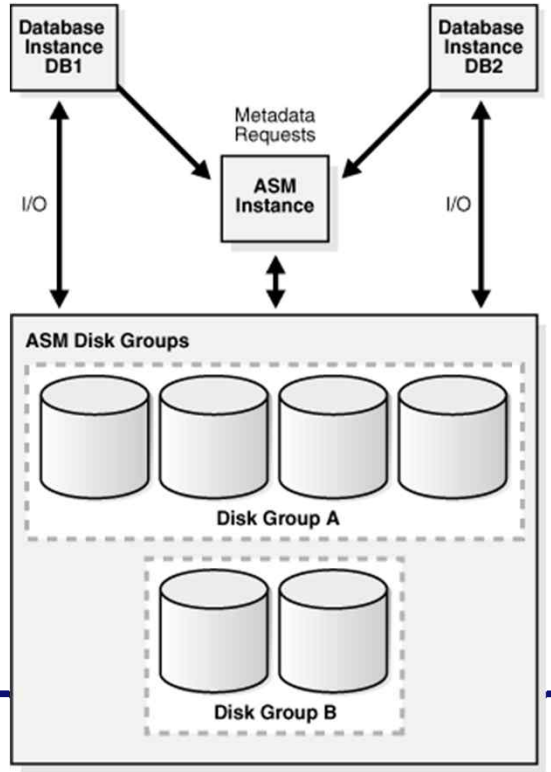


Database is essentially data storage system

- Oracle Physical storage structure



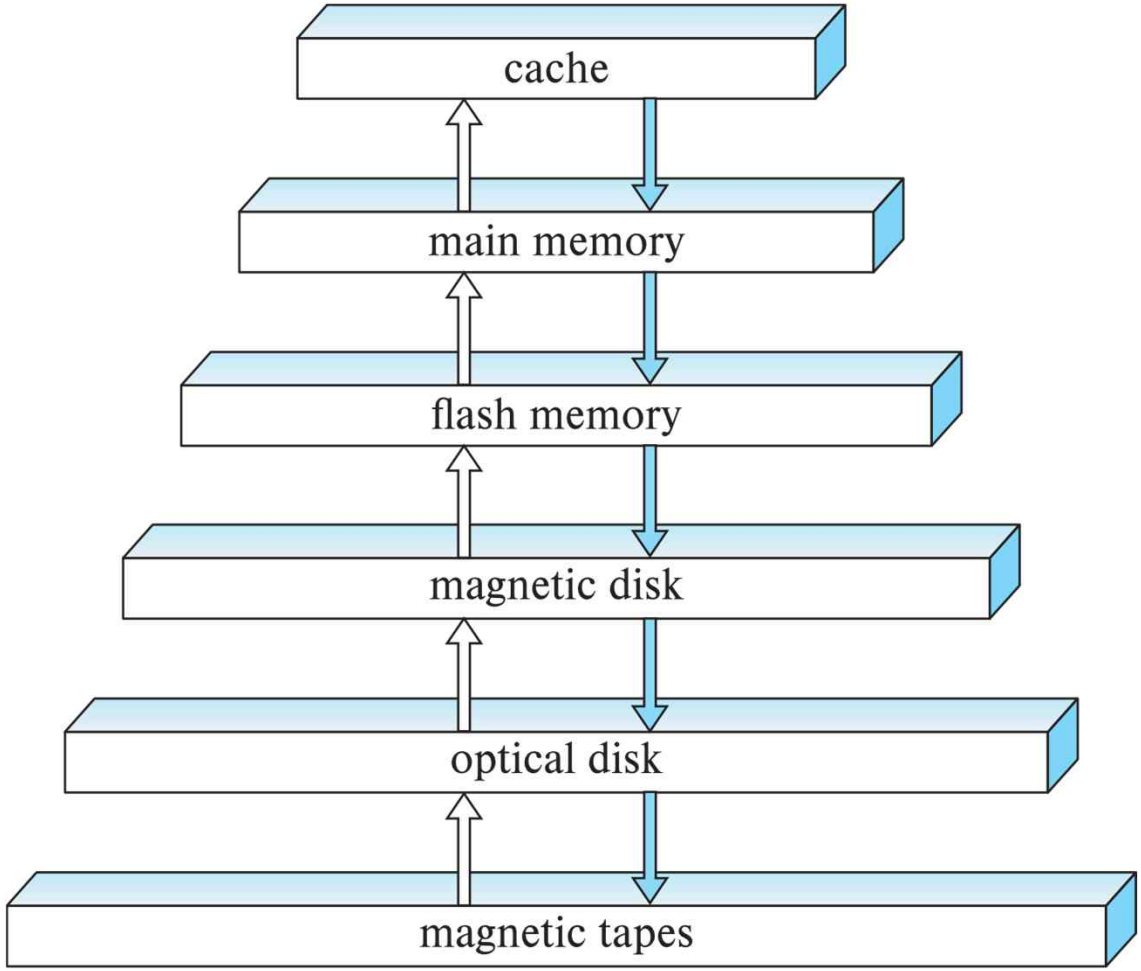




Classification of Physical Storage Media

- Can differentiate storage into:
  - **Volatile storage:** loses contents when power is switched off
  - **Non-volatile storage**:
    - Contents persist even when power is switched off
    - Includes secondary and tertiary storage, as well as battery-b acked up main-memory
- Factors affecting choice of storage media include
  - Speed with which data can be accessed
  - Cost per unit of data
  - Reliability

Storage Hierarchy



Storage Hierarchy

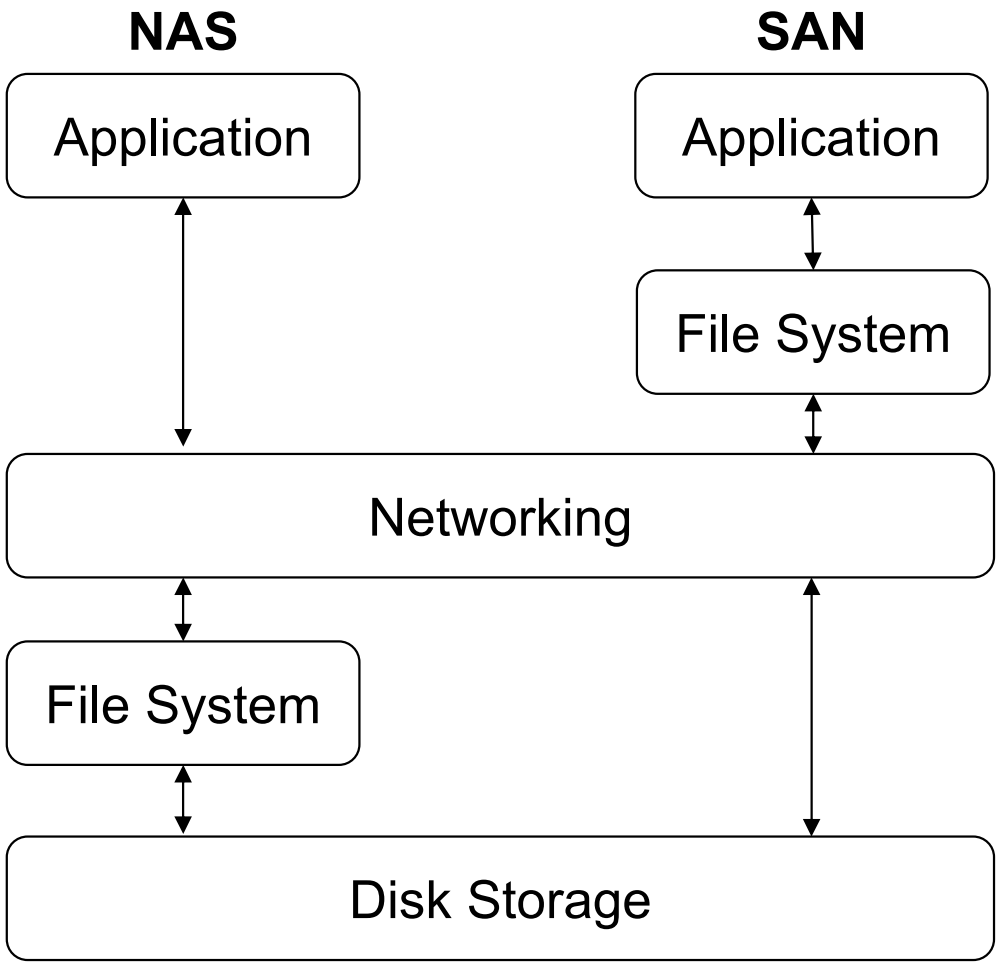
- **Primary storage**
  - Fastest media but volatile (cache, main memory)
- **Secondary storage**
  - Next level in hierarchy, non-volatile, moderately fast access time
    - Also called on-line storage
    - E.g., flash memory, magnetic disks
- **Teriary storage**
  - Lowest level in hierarchy, non-volatile, slow access time
  - E.g., magnetic tape, optical storage

Storage Interfaces

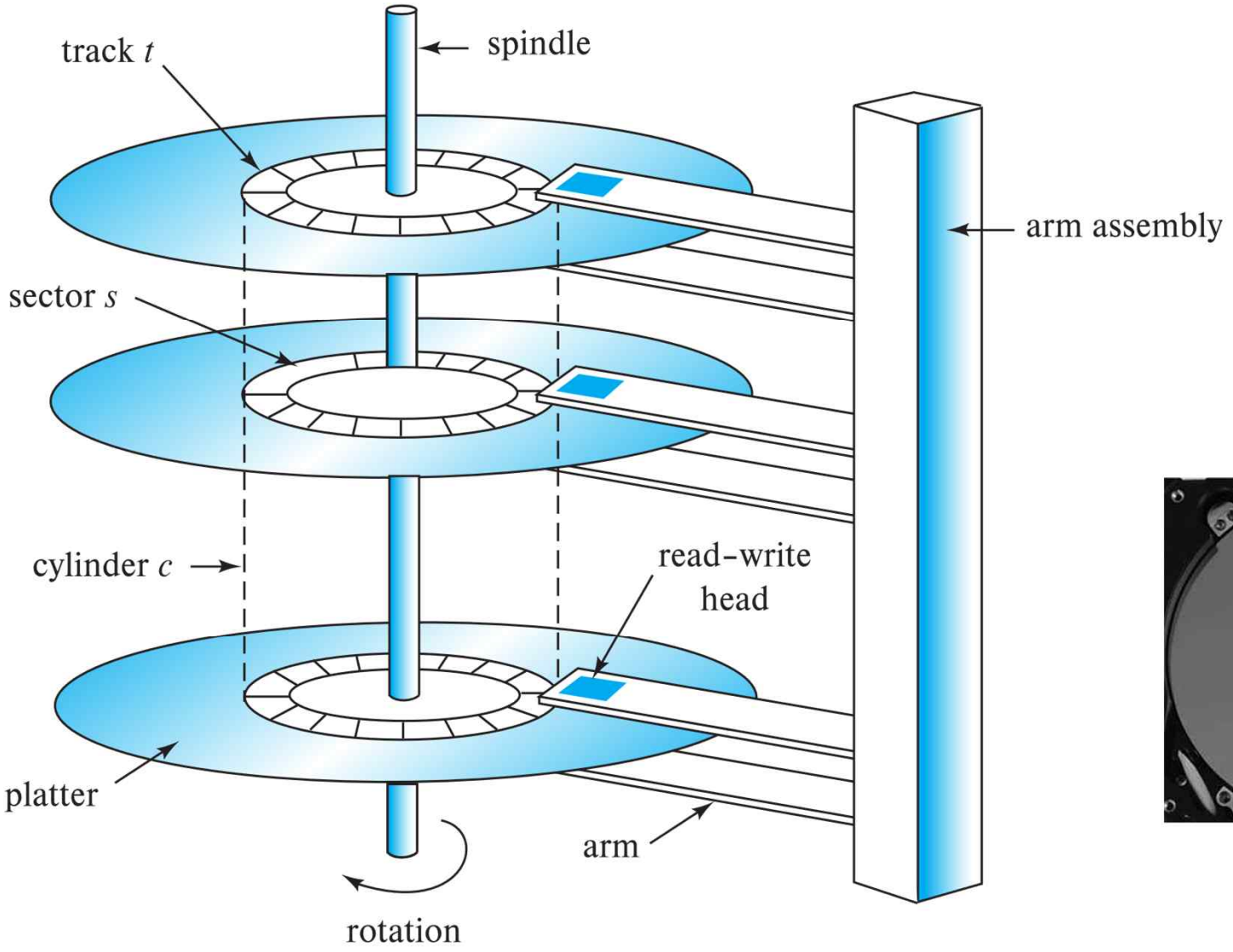
- Storage interface standards
  - Storage usually connected directly to computer system
    - **SATA** (Serial ATA)
      - SATA 3 supports data transfer speeds of up to 6 gigabits/sec
    - **SAS** (Serial Attached SCSI)
      - SAS Version 3 supports 12 gigabits/sec
    - **NVMe** (Non-Volatile Memory Express) interface
      - Works with PCIe connectors to support lower latency and higher transfer rates
      - Supports data transfer rates of up to 24 gigabits/sec
  - In storage area networks (**SAN**), a large number of disks are connected by a high-speed network to a number of servers
  - In network attached storage (**NAS**) networked storage provides a file system interface using networked file system protocol, instead of providing a disk system interface

Storage Interfaces

- **Comparison of NAS and SAN**



Magnetic Hard Disk Mechanism



Schematic diagram of magnetic disk drive

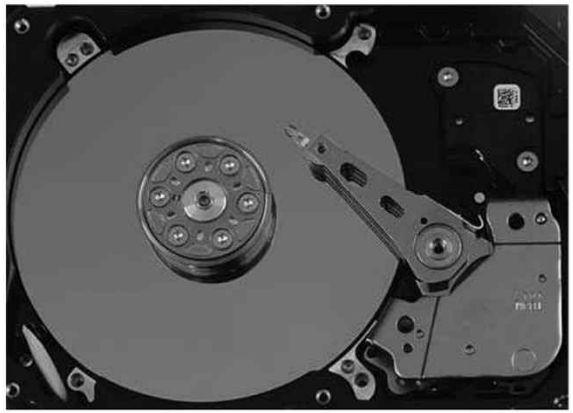


Photo of magnetic disk drive

Magnetic Disks

- **Read-Write head**
- Surface of platter divided into circular **tracks**
- Each track is divided into **sectors**
  - A sector is the smallest unit of data that can be read or written
  - Sector size typically 512 bytes
  - Typical sectors per track
    - 500 to 1000 (on inner tracks) to 1000 to 2000 (on outer tracks)
- To read/write a sector
  - Disk arm swings to position head on right track
  - Platter spins continually; data is read/written as sector passes under head
- Head-disk assemblies
  - Multiple disk platters on a single spindle (1 to 5 usually)
  - One head per platter, mounted on a common arm
- Cylinder i consists of i th track of all the platters

Magnetic Disks

- **Disk controller**
  - Interfaces between the computer system and the disk drive hardware
    - Accepts commands to read or write a sector
    - Initiates actions such as moving the disk arm to the right track and actually reading or writing the data
    - Ensures successful writing by reading back sector after writing it
    - Performs remapping of bad sectors

Performance Measures of Disks

- **Access time**
  - The time it takes from when a read or write request is issued to when data transfer begins
  - Seek time
    - Time it takes to reposition the arm over the correct track
      - 4 to 10 milliseconds on typical disks
  - Rotational latency
    - Time it takes for the sector to be accessed to appear under the head
      - 4 to 11 milliseconds on typical disks
  - Overall latency is 5 to 20 msec depending on disk model
- **Data-transfer rate**
  - The rate at which data can be retrieved from or stored to the disk
    - 25 to 200 MB per second max rate

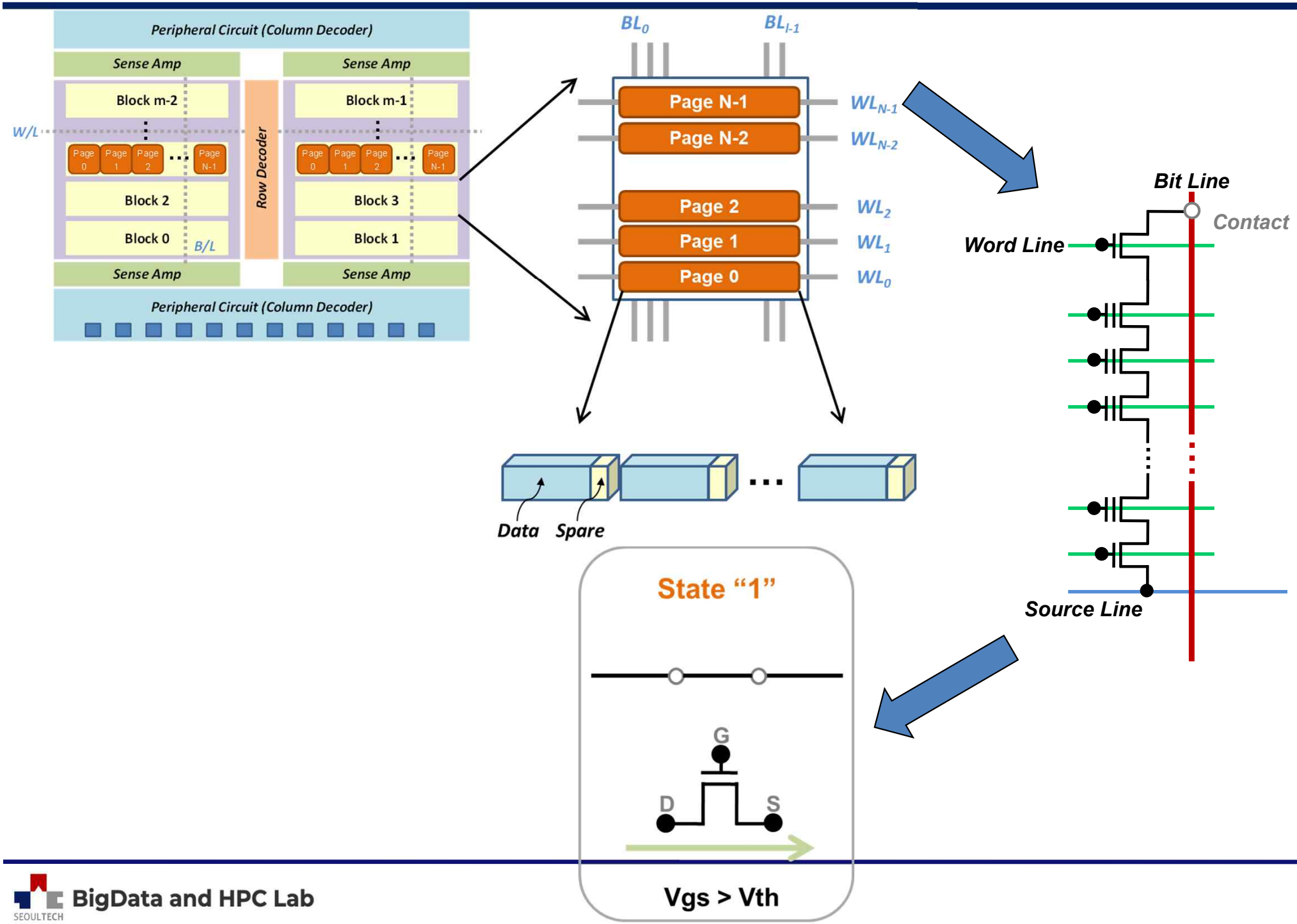
Performance Measures

- **Disk block** is a logical unit for storage allocation and retrieval
  - 4 to 16 kilobytes typically
    - Smaller blocks: more transfers from disk
    - Larger blocks: more space wasted due to partially filled blocks
- **Sequential access pattern**
  - Successive requests are for successive disk blocks
  - Disk seek required only for first block
- **Random access pattern**
  - Successive requests are for blocks that can be anywhere on disk
  - Each access requires a seek
  - Transfer rates are low since a lot of time is wasted in seeks
- I/O operations per second (IOPS)
  - Number of random block reads that a disk can support per second
  - 50 to 200 IOPS on current generation magnetic disks

Flash Storage

- **NAND flash**
  - Used widely for storage
  - Requires page-at-a-time read (page: 4KB or 8KB)
    - 20 to 100 microseconds for a page read
    - Not much difference between sequential and random read
  - Page can only be written once
    - Must be erased to allow rewrite
- **Solid state disks**
  - A solid-state drive (SSD) is a solid-state storage device that uses integrated circuit assemblies to store data persistently, typically using flash memory, and functioning as secondary storage in the hierarchy of computer storage.
  - Transfer rate of up to 500 MB/sec using SATA, and up to 3 GB/sec using NVMe PCIe

NAND flash Overview

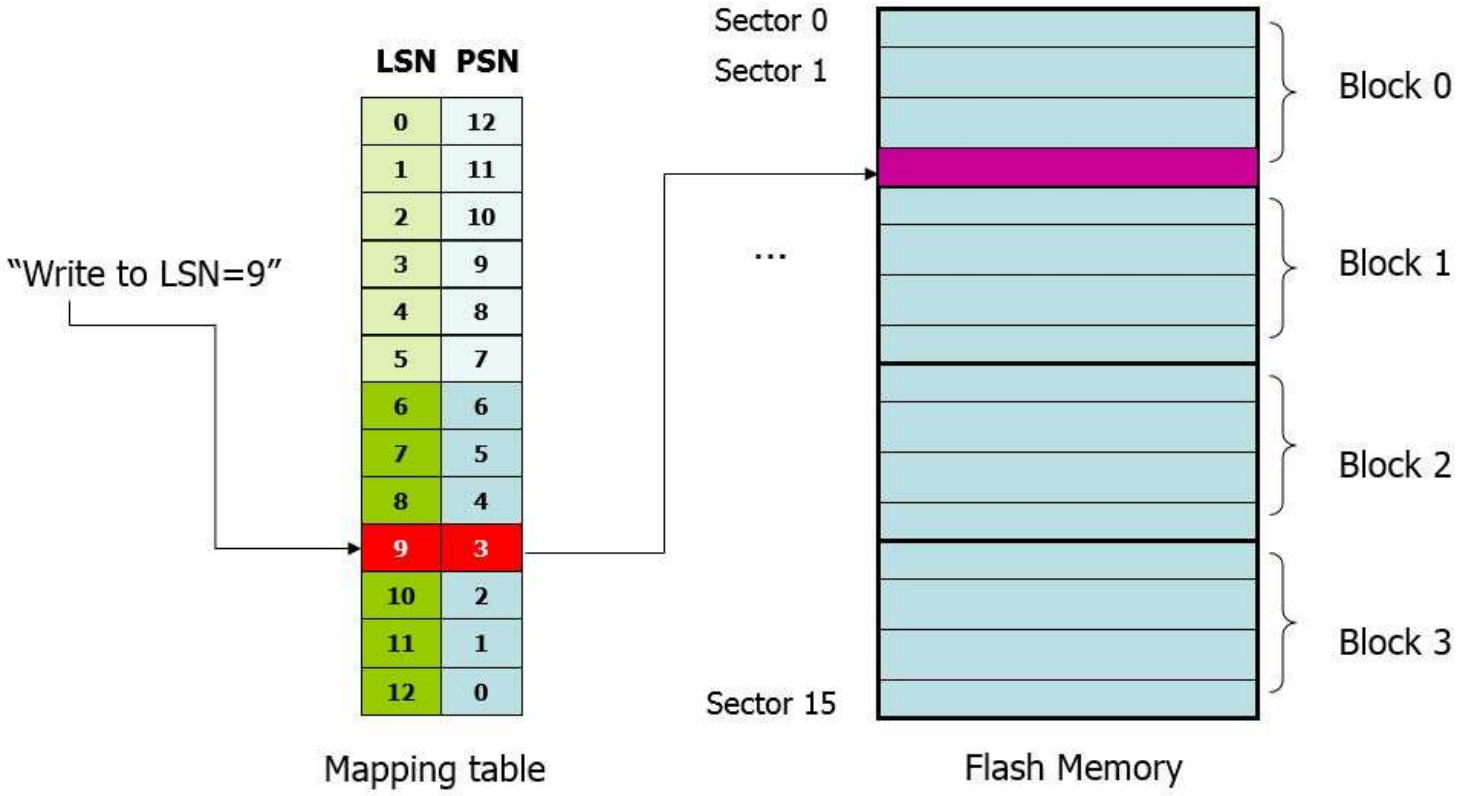


Flash Storage

- Erase happens in units of erase block
  - Takes 2 to 5 millisecs
  - Erase block typically 256 KB to 1MB
- **Remapping** of logical page addresses to physical page addresses avoids waiting for erase
- **Flash translation table** tracks mapping
  - Also stored in a label field of flash page
  - Remapping carried out by flash translation layer
- After 100,000 to 1,000,000 erases, erase block becomes unreliable and cannot be used
  - **Wear leveling**

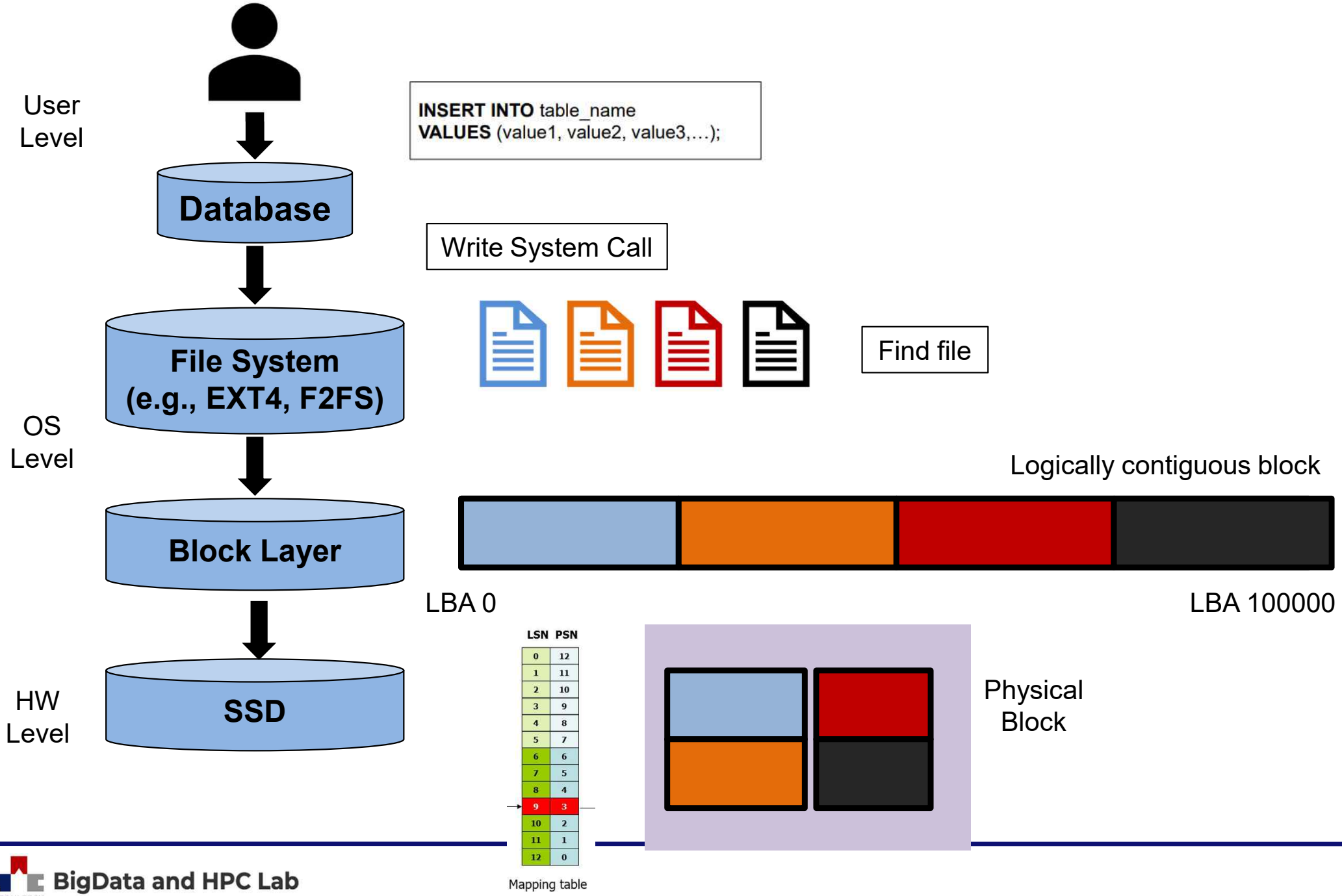
Flash Storage

- An example of remapping
  - Sector Mapping



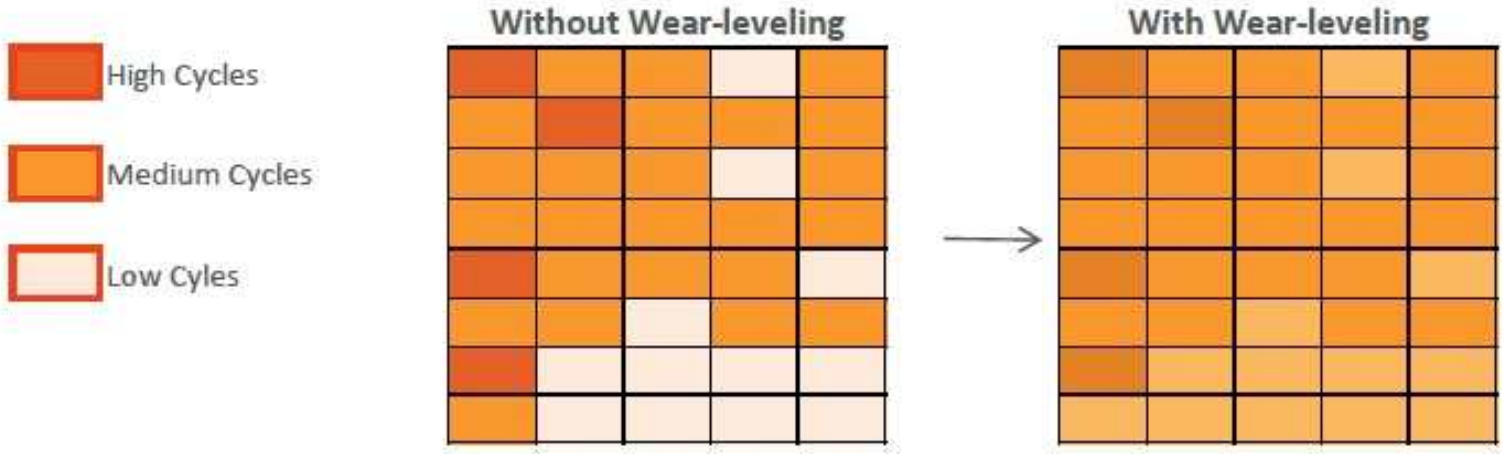
LSN: Logical Sector Number
PSN: Physical Sector Number

Database to Flash memory



Flash Storage

- **Wear leveling**
  - It is a technique for prolonging the service life of erasable storage media such as flash memory (SSDs)



  - Wear leveling takes data from blocks with many writes, redistributing it to blocks with few writes.
  - In theory, this will extend the disk lifetime by spreading writes around the disk.

Storage Class Memory

- 3D-Xpoint memory technology pioneered by Intel
  - 3D Xpoint is a non-volatile memory (NVM) technology
- Available as Intel Optane
  - SSD interface shipped from 2017
    - Allows lower latency than flash SSDs
  - Non-volatile memory interface announced in 2018
    - Supports direct access to words, at speeds comparable to main-memory speeds

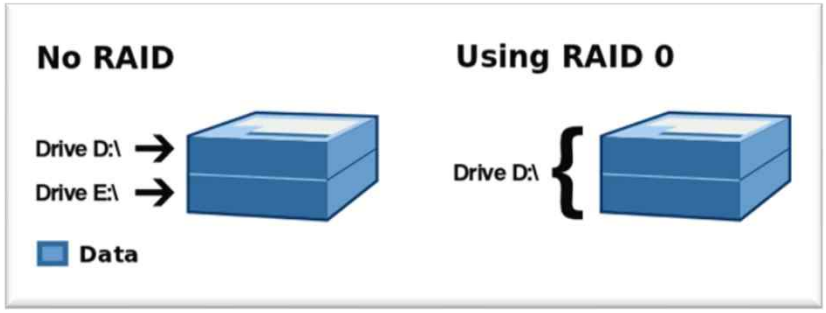


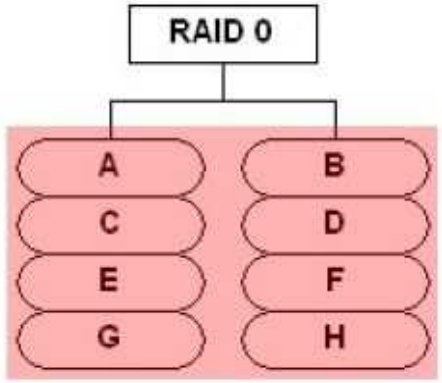
RAID

- **RAID: Redundant Arrays of Independent Disks**
  - Disk organization techniques that manage a large number of disks providing a view of a single disk of
    - High capacity and high speed by using multiple disks in parallel
    - High reliability by storing data redundantly, so that data can be recovered even if a disk fails
- The chance that some disk out of a set of N disks will fail is much higher than the chance that a specific single disk will fail
  - Techniques for using redundancy to avoid data loss are critical with large number of disks

RAID

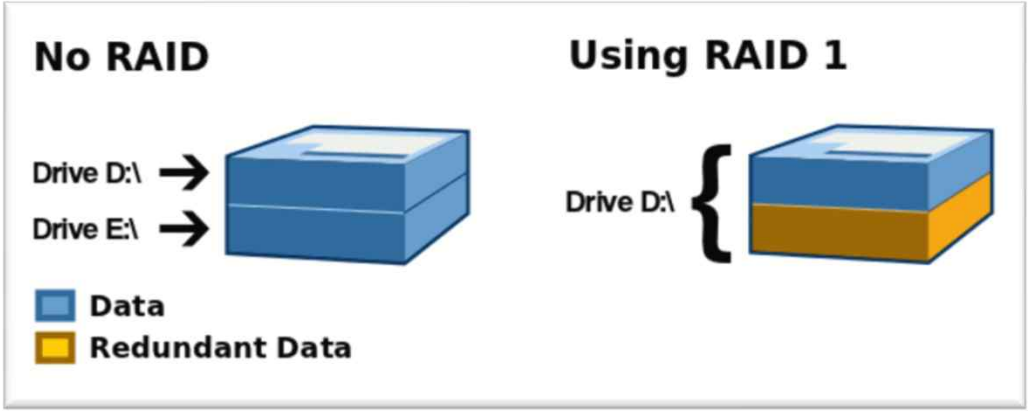
- **RAID 0:** block striping; non-redundant
  - With RAID0, we have two or more drives acting as if they were one single drive.
  - However, RAID0 offers no protection
  - **Advantages:**
    - RAID0 uses hard drive space to it maximum, since it has no redundancy
    - High performance
  - **Disadvantages:**
    - No protection

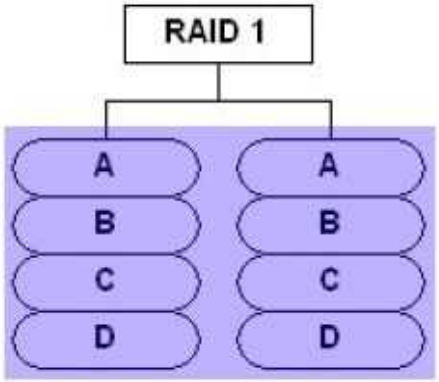




RAID

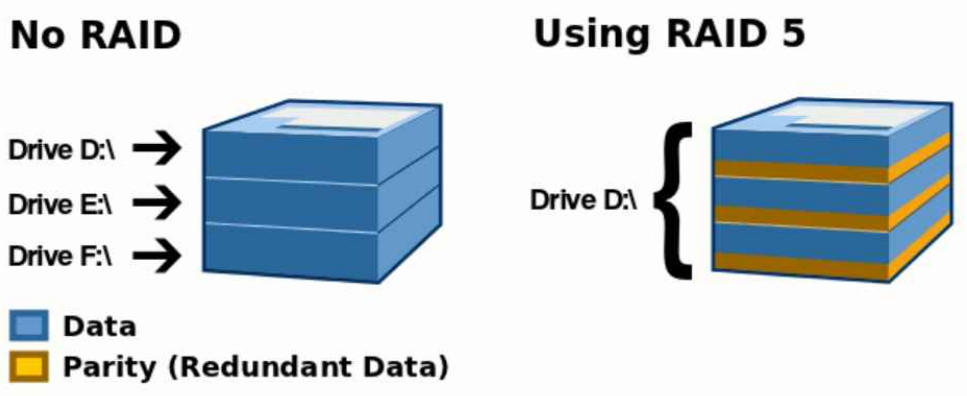
- **RAID 1:** Mirrored disks with block striping
  - In RAID 1, hard drives are mirrors of each other
    - This provides redundancy
  - Unlike RAID 0 where the total combined space of the drives is used, RAID 1 only uses half of the space, since the second drive is used for redundancy
    - Both hard drives have to be of equal sizes
  - **Advantages:**
    - Redundancy
  - **Disadvantages**:
    - Space is not used efficiently

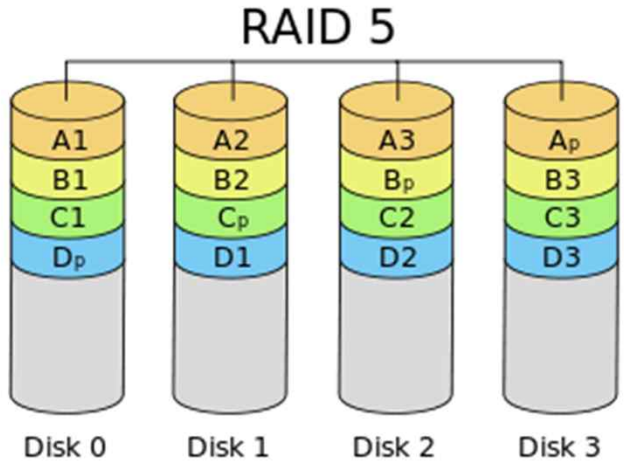




RAID

- **RAID 5: Block-interleaved Distributed Parity**
  - RAID5 consists of block-level striping with distributed parity
  - Upon failure of a single drive, subsequent reads can be calculated from the distributed parity such that no data is lost
    - RAID 5 requires at least three disks
  - Partitions data and parity among all disks





RAID

- **RAID 5: Block-interleaved Distributed Parity**
  - Reconstructing the lost data in a RAID 5 is quite simple:
    - We just perform a XOR operation between the still remaining data and the parity information, because:

A XOR B = Parity
A XOR Parity = B
B XOR Parity = A

Improvement of Reliability via Redundancy

- **Redundancy**
  - It store extra information that can be used to rebuild information lost in a disk failure
    - E.g., Mirroring or shadowing
      - Duplicate every disk
        - Logical disk consists of two physical disks
      - Every write is carried out on both disks
        - Reads can taks place from either disk
      - If one disk in a pair fails, data still available in the other
        - Data loss would occur only if a disk fails, and its mirror disk also fails before the system is repaired
          - » Probability of combined event is very small
          - » Except for dependent failure modes such as fire or building collapse or electrical power surges

RAID Types

- **There are two ways RAID can be implemented:**
  - Software RAID
    - In “Software RAID”, all the configuration is handled by the operating system.
      - RAID implementations done entirely in software, with no special hardware support
      - The **advantage** of software RAID is that it is very cheap, since you do not have to buy extra hardware and the software to manage the RAID is included with most operating systems
      - One **disadvantage** is that since it is not managed by it own hardware, it takes resources away from the computer
  - Hardware RAID
    - In “Hardware RAID”, all information regarding your specific RAID configuration is handled by a hardware interface card
      - The **advantage** of hardware RAID is that everything is handled by the interface card, the computer does no have to use memory or CPU power to run the RAID

Optimization of Disk-Block Access

- **Buffering**
  - In-memory buffer to cache disk blocks
- **Read-ahead**
  - Read extra blocks from a track in anticipation that they will be requested soon
- **Disk-arm-scheduling**
  - Scheduling algorithms re-order block requests so that disk arm movement is minimized
  - Elevator algorithm
    - The following is an example of how to calculate average disk seek times
      - Example list of pending disk requests (listed by track number): 100, 50, 10, 20, 75.
      - The list will need to be sorted in ascending order: 10, 20, 50, 75, 100.

Optimization of Disk Block Access

- **File organization**
  - Allocate blocks of a file in as contiguous a manner as possible
  - Files may get fragmented
    - E.g., if free blocks on disk are scattered, and newly created file has its blocks scattered over the disk
    - Sequential access to a fragmented file results in increased disk arm movement
    - Some systems have utilities to defragment the file system, in order to speed up file access